Gas, totals and mass flow

Cold gas (stations 0-3, 13-19): γ_c = 1.4, cp_c = 1004.5 J/(kg K). Hot gas (4-9): γ_t = 1.3, cp_t = 1239 J/(kg K). R = 287 J/(kg K) for air.

$$P = \rho R T, \quad R = c_p \frac{\gamma - 1}{\gamma}, \quad a = \sqrt{\gamma R T}, \quad \rho_1 v_1 A_1 = \rho_2 v_2 A_2$$

$$\dot{Q} - \dot{W} = \dot{m}\left[\left(h + \tfrac{v^2}{2}\right)_{out} - \left(h + \tfrac{v^2}{2}\right)_{in}\right]$$

W > 0 extracted (turbine), W < 0 delivered (compressor)

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}, \qquad \left.\frac{T_2}{T_1}\right|_s = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_t = T + \frac{v^2}{2c_p}, \quad \frac{T_t}{T} = 1 + \frac{\gamma-1}{2}M^2, \quad \frac{P_t}{P} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{P_{t2}}{P_{t1}} = e^{-\Delta s/R}$$

adiabatic inlet: $\tau_d = 1$ but $\pi_d < 1$; normal shock keeps Tt, loses Pt

$$\dot{m} = \frac{P_t A\,\mathrm{MFP}(M)}{\sqrt{R T_t}}, \qquad \mathrm{MFP} = \sqrt{\gamma} M \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

MFP max at M = 1: 0.6673 (γ 1.3), 0.6847 (γ 1.4); oblique flow: A cos(α)

Cycle notation and components

$$\tau_r = 1 + \frac{\gamma_c - 1}{2}M_0^2, \quad \pi_r = \tau_r^{\frac{\gamma_c}{\gamma_c - 1}}, \quad \tau_\lambda = \frac{c_{pt} T_{t4}}{c_{pc} T_0}, \quad f = \frac{\dot{m}_f}{\dot{m}_3}$$

$$\pi_d = \pi_{d,\max}\eta_r, \quad \eta_r = 1\ (M_0 \le 1),\ 1 - 0.075(M_0 - 1)^{1.35}\ (1 < M_0 < 5),$$

$$\eta_d = \frac{(P_{t2}/P_0)^{\frac{\gamma-1}{\gamma}} - 1}{T_{t0}/T_0 - 1}$$

$$\tau_c = \pi_c^{\frac{\gamma_c - 1}{\gamma_c e_c}} = 1 + \frac{\pi_c^{(\gamma_c-1)/\gamma_c} - 1}{\eta_c}, \qquad \dot{W}_c = \dot{m} c_{pc}(T_{t3} - T_{t2})$$

$$\varepsilon = \frac{\gamma - 1}{\gamma}: \quad \eta_c = \frac{\pi_c^\varepsilon - 1}{\tau_c - 1},\ e_c = \frac{\varepsilon \ln \pi_c}{\ln \tau_c}; \qquad \eta_t = \frac{1 - \tau_t}{1 - \pi_t^\varepsilon},\ e_t = \frac{\ln \tau_t}{\varepsilon \ln \pi_t}$$

e divides for a compressor (τ = π^(ε/e)), multiplies for a turbine (τ = π^(ε e))

$$f = \frac{c_{pt} T_{t4} - c_{pc} T_{t3}}{\eta_b h_{PR} - c_{pt} T_{t4}} = \frac{\tau_\lambda - \tau_r \tau_c}{\eta_b h_{PR}/(c_{pc} T_0) - \tau_\lambda}, \qquad \pi_b \approx 1 - \varepsilon\frac{\gamma}{2}M_b^2$$

exercises: π_b = 0.90, h_PR = 42.5 MJ/kg; stoichiometric f ≈ 0.068

$$\tau_t = 1 - \frac{\tau_r(\tau_c - 1)}{\eta_m(1+f)\tau_\lambda}, \qquad \pi_t = \tau_t^{\frac{\gamma_t}{(\gamma_t-1)e_t}} = \left[1 - \frac{1 - \tau_t}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t - 1}}$$

$$\frac{P_t}{P^*} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}} = 1.832\ (1.3),\ 1.893\ (1.4);$$

$$T^* = \frac{2T_t}{\gamma + 1}, \quad P^* = P_t\left[1 - \frac{1 - T^*/T_t}{\eta_n}\right]^{\frac{\gamma}{\gamma-1}}$$

P* ≥ P0 → choked, P9 = P*, M9 = 1; else matched P9 = P0, M9 < 1

Thrust and performance

$$F = \dot{m}_9 v_9 - \dot{m}_0 v_0 + (P_9 - P_0)A_9,$$

$$A_9 = \frac{\dot{m}_9 R T_9}{P_9 v_9}, \quad I = PA(1 + \gamma M^2)$$

two streams: add $\dot{m}_{19}$ v19 + (P19 - P0) A19

$$\mathrm{TSFC} = \frac{\dot{m}_f}{F}, \quad \eta_T = \frac{\Delta E_k}{\dot{m}_f h_{PR}},$$

$$\eta_P = \frac{F v_0}{\Delta E_k}, \quad \eta_O = \eta_T \eta_P = \frac{v_0}{h_{PR}\,\mathrm{TSFC}}$$

$$\Delta E_k = \tfrac{1}{2}\left(\dot{m}_9 v_9^2 + \dot{m}_{19} v_{19}^2 - \dot{m}_0 v_0^2\right), \qquad \eta_P \approx \frac{2}{1 + v_9/v_0}$$

Isp = F/($\dot{m}_f$ g); η_P with the FULL thrust (pressure terms included)

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r\pi_d\pi_c\pi_b\pi_t\pi_n, \qquad M_9 = \sqrt{\frac{2}{\gamma_t - 1}\left[\left(\frac{P_{t9}}{P_9}\right)^{\frac{\gamma_t - 1}{\gamma_t}} - 1\right]}$$

$$\frac{T_9}{T_0} = \frac{\tau_\lambda \tau_t c_{pc}/c_{pt}}{(P_{t9}/P_9)^{(\gamma_t-1)/\gamma_t}}, \qquad \frac{v_9}{a_0} = M_9\sqrt{\frac{\gamma_t R_t T_9}{\gamma_c R_c T_0}}$$

$$\frac{F}{\dot{m}_0} = a_0\left[(1+f)\frac{v_9}{a_0} - M_0 + (1+f)\frac{R_t}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1 - P_0/P_9}{\gamma_c}\right],$$

$$\mathrm{TSFC} = \frac{f}{F/\dot{m}_0}$$

Turbojet in 8 steps: (1) a0, v0, τ_r, π_r (2) η_r, π_d (3) τ_c (4) τ_{lambda}, f (5) τ_t from the power balance, then π_t (6) nozzle: Pt9/P9 → M9, choked check (7) T9/T0, v9/a0 (8) F/$\dot{m}_0$, TSFC, efficiencies.

$$\tau_{c,\mathrm{opt}} = \frac{\sqrt{\tau_\lambda}}{\tau_r}$$

Turbofan and two spools

$$\alpha = \frac{\dot{m}_{sec}}{\dot{m}_{core}}, \quad \dot{m}_{core} = \frac{\dot{m}_0}{1 + \alpha},$$

$$\dot{m}_f = f\dot{m}_{core}, \quad \mathrm{TSFC} = \frac{f}{(1+\alpha)F/\dot{m}_0}$$

$$\dot{W}_{HPC} = \eta_{m,HP}(1+f)\dot{m}_{core} c_{pt}(T_{t4} - T_{t45})$$

$$\dot{W}_{fan} + \dot{W}_{LPC} = \eta_{m,LP}(1+f)\dot{m}_{core} c_{pt}(T_{t45} - T_{t5}),$$

$$\dot{W}_{fan} = \dot{m}_0 c_{pc}(T_{t13} - T_{t2}), \quad \dot{W}_{LPC} = \dot{m}_{core} c_{pc}(T_{t25} - T_{t13})$$

$$\mathrm{OPR} = \pi_f \pi_{LPC} \pi_{HPC},$$

$$T_{t,out} = T_{t,in}\left[1 + \frac{\pi^\varepsilon - 1}{\eta}\right], \quad T_{t0} = T_0(1 + 0.2M_0^2)$$

$$F = \dot{m}_9 v_9 + \dot{m}_{19} v_{19} - \dot{m}_0 v_0 + (P_9 - P_0)A_9 + (P_{19} - P_0)A_{19},$$

$$FR = \frac{F_{core}/\dot{m}_{core}}{F_{sec}/\dot{m}_{sec}}$$

Order: compressions → f → HP balance (Tt45) → LP balance (Tt5) → π from η → nozzle checks → F, TSFC, efficiencies. Mixed exhaust: Pt6 = Pt16 fixes π_t, α becomes an output.

Turboshaft: PSFC = $\dot{m}_f$ / W_shaft.

Propellers, turboprop, open rotor

$$v_p = \frac{v_0 + v_e}{2}, \quad a = \frac{v_p - v_0}{v_0}, \quad F = 2\dot{m} a v_0,$$

$$\dot{m} = \rho_0 A_D v_p, \quad \dot{W}_D = F v_p, \quad \eta_P = \frac{1}{1 + a}$$

$$v_p = \frac{v_0 + \sqrt{v_0^2 + 2F/(\rho_0 A_D)}}{2},$$

$$\eta_{prop} = \eta_L \eta_P, \quad \eta_L = \eta_{prop}(1 + a) \le 1$$

$$\eta_{prop} = 10 M_0 \eta_{\max}\ (M_0 \le 0.1), \quad \eta_{\max}\ (0.1 < M_0 \le 0.7),$$

$$\eta_{\max}\left[1 - \frac{M_0 - 0.7}{3}\right]\ (0.7 < M_0 \le 0.85)$$

$$F_{static} = (2\rho_0 A_D)^{1/3}(F M \dot{W}_{prop})^{2/3}, \qquad FM \approx 0.70\text{-}0.80$$

$$\dot{W}_{prop} = \eta_g(\eta_{m,L}\dot{W}_{LPT} - \dot{W}_{LPC}),$$

$$F_{prop} = \frac{\eta_{prop}\dot{W}_{prop}}{v_0}, \quad \mathrm{PSFC} = \frac{\dot{m}_f}{\dot{W}_{prop}}$$

$\Delta E_k = \frac{1}{2}\dot{m}_9 v_9^2 - \frac{1}{2}\dot{m}_{jet}v_0^2 + \dot{W}_D$, $A_{disc} = \pi\frac{D^2}{4} - \dot{m}_{45}\frac{\sqrt{RT_{t45}}}{P_{t45}\,\mathrm{MFP}_{45}}$

Turbomachinery: Euler and conventions

$$\tau_A = \dot{m}(r_e v_e - r_i v_i), \qquad h_{te} - h_{ti} = \omega(r_e v_e - r_i v_i)$$

$$\text{compressor: } c_p\Delta T_t = U(v_2 - v_1),$$

$$\text{turbine: } c_p\Delta T_t = U(v_2 + v_3), \quad U = \omega r_m$$

$$\vec{V} = \vec{V}_R + \vec{U}, \quad v = u\tan\alpha, \quad v_R = u\tan\beta$$

compressor: rotor 1-2 then stator 2-3, v + v_R = U; turbine: nozzle 1-2 then rotor 2-3, v2R = v2 - U, v3R = v3 + U (v3 positive AGAINST U)

P, T, ρ are frame-invariant; Tt, Pt, M are not. Rotor: Tt,R constant (same radius). Stator: Tt constant.

Compressor stage

$$c_p\Delta T_t = Uu(\tan\beta_1 - \tan\beta_2) = Uu(\tan\alpha_2 - \tan\alpha_1)$$

$$T_{t1R} = T_1 + \frac{V_{1R}^2}{2c_p} = T_{t2R}, \quad P_{t2R} = L_R P_{t1R},$$

$$P_{t3} = L_S P_{t2}, \quad L_R = 1 - \phi_{cr}\frac{\gamma P_1 M_{1R}^2}{2P_{t1R}}$$

$$R_c = \frac{T_2 - T_1}{T_3 - T_1}, \quad \Psi = \frac{c_p\Delta T_t}{U^2}, \quad \Phi = \frac{u}{U}, \quad \eta_{cs} = \frac{\pi_{cs}^{\varepsilon} - 1}{\tau_{cs} - 1}$$

$$D = 1 - \frac{V_e}{V_i} + \frac{|v_i - v_e|}{2\sigma V_i} \le 0.6$$

$$\sigma = \frac{c}{s}, \quad s = \frac{2\pi r_m}{n_b}, \quad i = \alpha_i - \gamma_i, \quad \delta = \alpha_e - \gamma_e,$$

$$\delta_c = \frac{\gamma_i - \gamma_e}{4\sqrt{\sigma}}, \quad \delta_t = \frac{\gamma_i + \gamma_e}{8\sqrt{\sigma}}$$

$$\text{repeating stage: } v_1 + v_2 = U, \quad c_p\Delta T_t = V_2^2 - V_1^2,$$

$$\frac{\omega r}{V_1} = \cos\alpha_1(\tan\alpha_1 + \tan\alpha_2)$$

$$\tau_{cs} = 1 + \frac{(\gamma-1)M_1^2}{1 + \frac{\gamma-1}{2}M_1^2}\left(\frac{\cos^2\alpha_1}{\cos^2\alpha_2} - 1\right),$$

$$\Psi = \frac{\tan\alpha_2 - \tan\alpha_1}{\tan\alpha_1 + \tan\alpha_2}, \quad \Phi = \frac{1}{\tan\alpha_1 + \tan\alpha_2}$$

$$\pi_c = \left[\frac{T_{t1} + n_s\Delta T_t}{T_{t1}}\right]^{\frac{\gamma e_c}{\gamma-1}}, \qquad n_s \approx \frac{T_{t1}\left(\pi_c^{(\gamma-1)/(\gamma e_c)} - 1\right)}{\Delta T_{t,\mathrm{stage}}}$$

stage ratios fall along the machine: $\pi_c \neq \pi_{cs}$^n

$\frac{dh_t}{dr} = u\frac{du}{dr} + v\frac{dv}{dr} + \frac{v^2}{r}$, free vortex: $R = 1 - (1 - R_m)\left(\frac{r_m}{r}\right)^2$

reaction lowest at the hub (turbine: keep R_t > -0.15 there)

$A = \frac{\dot{m}\sqrt{RT_t}}{P_t\cos\alpha\,\mathrm{MFP}(M)}$, $r_t = \sqrt{\frac{A}{\pi\left[1 - (r_h/r_t)^2\right]}}$, $h = \frac{A}{2\pi r_m}$

hub/tip 0.60-0.75 at HPC inlet; Pt in Pa

Turbine stage

$$T_{t2} = T_{t1}, \quad P_{t2} = L_S P_{t1}, \quad T_{t3R} = T_{t2R}, \quad P_{t3R} = L_R P_{t2R},$$

$$L = \frac{1}{1 + \phi_t\left[1 - (T/T_t)^{\frac{\gamma}{\gamma-1}}\right]}$$

$$\Psi = \frac{c_p\Delta T_t}{U^2}, \quad \Phi = \frac{u_2}{U},$$

$$\frac{\Psi}{\Phi} = \tan\beta_2 + \tan\beta_3, \quad \frac{R_t}{\Phi} = \frac{\tan\beta_3 - \tan\beta_2}{2}$$

$$R_t = \frac{T_2 - T_3}{T_{t1} - T_{t3}}, \qquad \text{zero exit swirl: } R_t = 1 - \frac{\Psi}{2}$$

0 % (impulse): $\beta_3 = \beta_2$, Ψ = 2, v2 = 2U; 50 %: $\alpha_2 = \beta_3$, $\alpha_3 = \beta_2$, Ψ = 1

$$\Phi = \frac{1}{\tan\alpha_2 - \tan\beta_2}, \qquad U = \sqrt{\frac{c_p\Delta T_t}{\Psi}}, \qquad u = \Phi U$$

map axes: turning $\beta_2 + \beta_3$ vs β_2; pick the LARGEST Ψ/Φ with $\alpha_2 \le 70°$ (and Zweifel)

$$\eta_{ts} = \frac{1 - T_{t3}/T_{t1}}{1 - (P_{t3}/P_{t1})^{(\gamma-1)/\gamma}}, \qquad Z_r = 2\frac{s}{c_x}\cos^2\beta_3(\tan\beta_2 + \tan\beta_3)$$

$$\sigma_c \approx \frac{\rho\omega^2 A}{4\pi}\left(1 + \frac{A_t}{A_h}\right) \Leftrightarrow AN^2 = \frac{3600\times10^3}{\pi(1 + A_t/A_h)}\frac{\sigma_c}{\rho}$$

N in rpm, A in m^2, σ_c/ρ in MPa/(g/cm^3) = 1000 m^2/s^2; too high: lower N, change taper, add a stage

Choking: first nozzle M2 > 1 and rotor exit M3R < 0.9. Sanity: r_m = U/ω and h = A/(2 π r_m) must be realistic; typical LPT U 350-460 m/s.